Akbar Elyom Acad.
CIS: First year
First Med.

• الخطأ، تحسابات
• الأرقام في الأسئلة أرقام عشرية
$\delta_o = O(1-O)(T-O)$

Akbar Elyom Academy CIS: fourth year First Med term Exam. nov.2025	أكاديمية أخبار اليوم AKHBAR ELYOM ACADEMY	Course : neural network Time allowed: 1 Hr. Grade=20 Examiner :Dr Hussein Omran

Question 1

8 1. Connections across layers in standard topologies & among the units within a layer organized?
   a) in feedforward manner b) in feedback manner
   c) both feedforward & feedback d) either feedforward & feedback

2. What is classification?
   a) deciding what features to use in a pattern recognition problem
   b) deciding what class an input pattern belongs to
   c) deciding what type of neural network to use
   d) none of the mentioned

3. What are models in neural networks?
   a) mathematical representation of our understanding
   b) representation of biological neural networks
   c) both way
   d) none of the mentioned

4. Can data be stored through weights?
   a) yes b) no.

5. What consist of competitive learning neural networks?
   a) feed forward paths b) feedback paths
   c) either feed forward or feedback d) combination of feed forward and feedback

6. How is weight vector adjusted in basic competitive learning?
   a) such that it moves towards the input vector b) such that it moves away from input vector
   c) such that it moves towards the output vector d) such that it moves away from output vector

7. What is capacity of a network?
   a) number of inputs it can take b) number of output it can deliver
   c) number of patterns that can be stored d) none of the mention

8. Number of desired patterns is what of basins of attraction?
   a) dependent b) independent c) dependent or independent d) none of the mentioned

d
b
c
a
d
d
c
b

Question 2

3 a) Explain why we prefer to use the logistic (sigmoid) function as the activation function in back-propagation networks. useful for binary classification & Probabilistic $(0 \div 1)$

3 b) What may happen when we use a very large learning rate, and when we use a very small learning rate? miss optimization if large & take much time if to small

6 Question3

gradient method

Given NN with weight w1=0.11,w2=0.21,w3=0.12,w4=0.08,w5=0.14,w6=0.15 and input with X1=10 , x2=20 ,target= 1, η=0.05 using sigmoid function to find the error and updated weight values then next output after update

$h_1 = 10 \times .11 + 20 \times .21 = 1.1 + 4.3 = 5.3 \rightarrow h_1(s) = .995$
$h_2 = 10 \times .12 + 20 \times .08 = 1.2 + 1.6 = 2.8 \rightarrow h_2(s) = .943$

$O = .14 \times .995 + .15 \times .943$
$O = .281 \rightarrow O(s) = .569$
error $= (T-O)(1-.569)$
error $= 0.430$ *

$\delta_o = O(1-O)(T-O) =$
$= .569 \times (1-.569)(1-.5697) = .1055$
$\therefore \boxed{\delta_o = .1055}$

$w_5 = w_5 + \eta \delta_o h_1 = .14 + .05 \times .1055 \times .995 = .145$

$w_6 = .15 + .05 \times .1055 \times .943 = 0.1550$

$\delta_{hi} = \delta_o h_i (1-h_i) \therefore \delta_{h1} = .1055 \times .995 \times (1-.995) = .995$
$\delta_{h2} = .1055 \times .943 \times (1-.943) = .9499$

$w_1 = w_1 + \eta \times \delta_{h1} X_1 = .11 + .05 \times .995 \times 10 = .11026$
$w_2 = .21 + .05 \times .995 \times 20 = .21052$

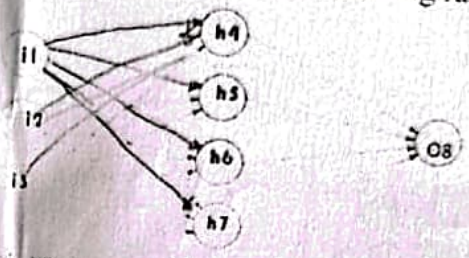
$h_1' = 5.313 = .995$
$h_2' = 2.942 = .943$
$O = .2917 \rightarrow .572$

$w_5' = .1452$
$w_6' = .1550$
$w_1' = .11026$
$w_2' = .21052$
$w_3' = .12285$
$w_4' = .0857$
error $= 0.4276$ *

Akhbar El yom Academy
CIS: second year
Second Term April .2025

Course name: Neural Network
Time allowed: 3Hr Total Grad=70
Examiner :Dr Hussein Omran page=1

1. Explain why we prefer to use the logistic (sigmoid) function as the activation function in back-propagation networks?
2. What do you mean by Backpropagation and why used?
3. What are the difference between different classifications
4. What Are the Different Layers on CNN?
5. Explain the different steps used in Gradient Descent Algorithmfor competitivelearning?
6. Suppose we have ANN with sigmoid activation (S) = 1/(1 + e-S) , Choose a learning rate η= 0.2) , start with inputs (i1=1,i2=2,i3=3) , desired outputs= 1 as a categorization problem
w14=w15=w16=w17=0.1.
w24=w25=w26=w27=0.2, and
w34=w35=w36=w37=0.3 and
w48=w58=w68=w78=0.4 Apply back propagation algorithm to find updated weights to reduce the errors.



7. given table of height, age and weight for 10 people. predict missed weight of person of 11 using KNN based on k=3 the K=5?

id	H	age	W
1	5	45	77
2	5.11	26	47
3	5.6	30	55
4	5.9	34	59
5	4.8	40	72
6	5.8	36	60
7	5.3	19	40
8	5.8	28	60
9	5.5	23	45
10	5.6	32	58
11	4.7	42	

8. For the given image and filter below apply convolution process and average padding to find the convolved image?

Image

9	9	0	0
9	9	0	0
9	9	0	0
9	9	0	0

filter

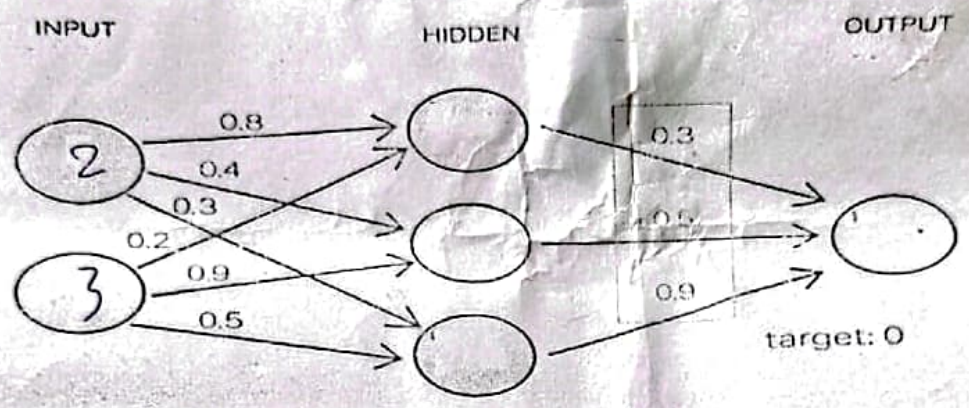
1	0	-1
1	0	-1
1	0	-1

Good luck

Akbar Elyom Academy CIS: fourth year Second Mid Term April .2025	أكاديمية أخبار اليوم AKHBAR ELYOM ACADEMY	Course : neural network Time allowed: 1 Hr. Grade=20 Examiner :Dr Hussein Omran

Solve following questions

1) State the different methods of neuron activations and function used? **(2 points)**
2) What are the difference between supervised and unsupervised **(2 points)**
3) How to train neural networks (state training Widrow-Hoff algorithm)? **(2 points)**
4) What may happen when learning rate very large, and when very small? **(2 points)**
5) For figure below, aapply back propagation algorithm to find the updated weights to reduce errors where $\sigma(S) = 1/(1 + e^{-S})$, and learning rate, $\eta = 0.1$, if given $w_1 = 0.8$, $w_2 = 0.4$, $w_3 = 0.3$, $w_4 = 0.2$, $w_5 = 0.9$, $w_6 = 0.5$, $w_7 = 0.3$, $w_8 = 0.5$, $w_9 = 0.9$ target=0, I1=2, I2=3?**(8 points)**



6: Multiple choice Select the correct answer (4 points)

1) What is classification?
   a) deciding what features to use in a pattern recognition problem
   b) deciding what class an input pattern belongs to
   c) deciding what type of neural network to use
   d) none of the mentioned

2) Which layer has feedback weights in competitive neural networks?
   a) input layer b) second layer c) both input and second layer d) none of mentioned

3) The update in weight vector in basic competitive learning can be represented by?
   a) $w(t + 1) = w(t) + \Delta w(t)$ b) $w(t + 1) = w(t)$
   c) $w(t + 1) = w(t) - \Delta w(t)$ d) none of the mentioned

4) What is capacity of a network?
   a) number of inputs it can take
   b) number of output it can deliver
   c) number of patterns that can be stored
   d) none of the mentioned

Good luck